



SYDNEY BOYS HIGH SCHOOL
MOORE PARK, SURRY HILLS

2014
HIGHER SCHOOL CERTIFICATE
ASSESSMENT TASK #1

Mathematics Extension 2

General Instructions

- Reading Time – 5 Minutes
- Working time – 90 Minutes
- Write using black or blue pen. Pencil may be used for diagrams.
- Board approved calculators may be used.
- Each question is to be returned in a separate bundle.
- All necessary working should be shown in every question.

Total Marks – 60

- Attempt questions 1 – 3
- All questions are of equal value.
- Unless otherwise directed give your answers in simplest exact form.

Examiner: *A.M.Gainford*

Question 1. (Start a new page.) (20 marks)

- | | Marks |
|---|--------------|
| (a) For the complex number $z = 1 - \sqrt{3}i$ find: | 3 |
| (i) $ z $ | |
| (ii) $\arg z$. | |
| (iii) $\frac{z}{i}$ | |
|
(b) Express the following in the form $a + ib$ (for real a and b). | 2 |
| (i) $(6 + 5i)\overline{(4 - i)}$ | |
| (ii) $\frac{-2 + 3i}{3 - 4i}$ | |
|
(c) Find the square roots of $9 + 40i$, giving your answers in the form $x + iy$. | 2 |

Question 1 continues on the next page.

- (d) Sketch (on separate diagrams) the region in the Argand diagram containing the points z for which: 4
- (i) $\frac{\pi}{4} \leq \arg(z) \leq \frac{\pi}{2}$ and $|z - 1 - 3i| \leq 2$
- (ii) $\arg\left(\frac{z - 2i}{z + 2}\right) = \frac{\pi}{4}$
- (e) (i) Express $1 + i$ in modulus-argument form. 1
- (ii) Given that $(1 + i)^n = x + iy$, where x and y are real, and n is an integer, prove that $x^2 + y^2 = 2^n$ 2
- (f) Which complex numbers are the reciprocals of their conjugates? 1
- (g) Consider the function $y = 2 \cos^{-1}(x^2 - 1)$. 5
- (i) Determine the domain and range of the function.
- (ii) Sketch the graph of the function showing important features.
- (iii) Find the derivative of the function and state the values of x for which it is defined.

Question 2. (Start a new page.) (20 marks)

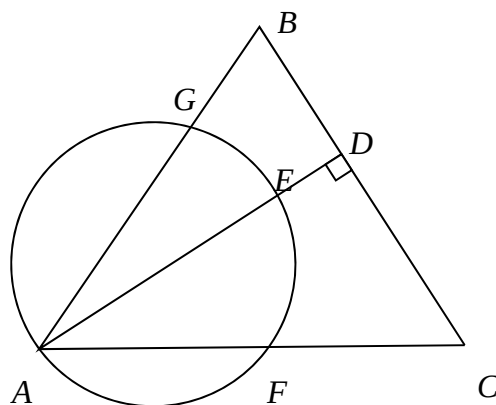
- | | Marks |
|--|--------------|
| (a) The points O, I, Z , and P on the Argand diagram represent the complex numbers $0, 1, z$, and $z+1$ respectively, where $z = \cos \theta + i \sin \theta$ is any complex number of modulus 1, and $0 < \theta < \pi$. | 4 |
| <div style="margin-left: 40px;">(i) Explain why $OIPZ$ is a rhombus.</div> <div style="margin-left: 40px;">(ii) Show that $\frac{z-1}{z+1}$ is purely imaginary.</div> <div style="margin-left: 40px;">(iii) Find the modulus of $z+1$ in terms of θ.</div> | |
| (b) Differentiate $x \sin 2x$, and hence find $\int x \cos 2x \, dx$. | 2 |
| (c) Given that $2-i$ is a root of the equation $x^4 - 6x^3 + 10x^2 + 2x - 15 = 0$: | 5 |
| <div style="margin-left: 40px;">(i) state another complex (non-real) root, giving a reason.</div> <div style="margin-left: 40px;">(ii) find all roots of the equation.</div> <div style="margin-left: 40px;">(iii) write the equation in fully factored form over the complex field.</div> | |
| (d) Consider the functions $y = -\cos^{-1}\left(\frac{x}{2}\right)$ and $y = \frac{1}{2}\tan^{-1}(x) - \frac{\pi}{2}$. | 4 |
| <div style="margin-left: 40px;">(i) Show that the graphs of these functions intersect on the y-axis.</div> <div style="margin-left: 40px;">(ii) Show that the graphs have a common tangent at the point of intersection, and write the equation of this tangent.</div> | |
| (e) Given the quadratic equation $x^2 - x - 3 = 0$ with roots α_1, α_2 : | 5 |
| <div style="margin-left: 40px;">(i) Show that $x^4 = 7x + 12$.</div> <div style="margin-left: 40px;">(ii) Hence or otherwise find a quadratic equation with roots α_1^4 and α_2^4.</div> | |

Question 3. (Start a new page.) (20 marks)

		Marks
(a)	(i) Find the five roots of the equation $z^5 = 1$. Give the roots in modulus-argument form.	2
	(ii) Show that $z^5 - 1$ can be factorised in the form : $z^5 - 1 = (z - 1)(z^2 - 2z \cos \frac{2\pi}{5} + 1)(z^2 - 2z \cos \frac{4\pi}{5} + 1)$	2
	(iii) Hence show that $\cos \frac{2\pi}{5} + \cos \frac{4\pi}{5} = -\frac{1}{2}$, and hence find the exact value of $\cos \frac{2\pi}{5}$.	3
(b)	When a polynomial $P(x)$ is divided by $x - 2$ and by $x - 3$ the remainders are 4 and 9 respectively. Find the remainder when $P(x)$ is divided by $(x - 2)(x - 3)$.	2
(c)	Ten people, consisting of three couples and four singles are to be seated randomly at a round table.	3
	(i) How many arrangements are possible?	
	(ii) What is the probability (as a simplified fraction) that all three couples are seated as couples, separated from other couples by one or two singles?	
(d)	Prove that the polynomial equation $ax^4 + bx + c = 0$, where a , b , and c are non-zero, cannot have a triple root.	1
(e)	Use the substitution $x = 2 \sin \theta$, or otherwise, to evaluate $\int_1^{\sqrt{3}} \frac{x^2}{\sqrt{4 - x^2}} dx$.	3

- (f) In the triangle ABC , AD is the perpendicular from A to BC . The point E is any point on AD , and the circle drawn with AE as diameter cuts AC at F and AB at G .

4



- (i) Copy the diagram to your answer booklet.
- (ii) Prove that B, G, F , and C are concyclic.

This is the end of the paper.

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STANDARD INTEGRALS

$$\int x^n dx = \frac{1}{n+1} x^{n+1}, \quad n \neq -1; \quad x \neq 0, \text{ if } n < 0$$

$$\int \frac{1}{x} dx = \ln x, \quad x > 0$$

$$\int e^{ax} dx = \frac{1}{a} e^{ax}, \quad a \neq 0$$

$$\int \cos ax dx = \frac{1}{a} \sin ax, \quad a \neq 0$$

$$\int \sin ax dx = -\frac{1}{a} \cos ax, \quad a \neq 0$$

$$\int \sec^2 ax dx = \frac{1}{a} \tan ax,$$

$$\int \sec ax \tan ax dx = \frac{1}{a} \sec ax, \quad a \neq 0$$

$$\int \frac{1}{a^2 + x^2} dx = \frac{1}{a} \tan^{-1} \frac{x}{a}, \quad a \neq 0$$

$$\int \frac{1}{\sqrt{a^2 - x^2}} dx = \sin^{-1} \frac{x}{a}, \quad a > 0, \quad -a < x < a$$

$$\int \frac{1}{\sqrt{x^2 - a^2}} dx = \ln \left(x + \sqrt{x^2 - a^2} \right), \quad x > a > 0$$

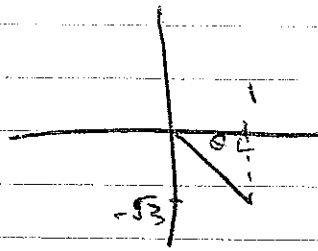
$$\int \frac{1}{\sqrt{x^2 + a^2}} dx = \ln \left(x + \sqrt{x^2 + a^2} \right)$$

NOTE: $\ln x = \log_e x, \quad x > 0$

i) a) i) $z = 1 - \sqrt{3}i$

$$|z| = \sqrt{(1)^2 + (-\sqrt{3})^2}$$

$$= 2$$



ii) $\arg z = -\frac{\pi}{3}$

iii) $\frac{1 - \sqrt{3}i}{i} \times \frac{-i}{-i} = \frac{-i - \sqrt{3}}{1}$

$$= -\sqrt{3} - i$$

b) i) $(6 + 5i)(\overline{4 - i}) = (6 + 5i)(4 + i)$

$$= 24 + 6i + 20i - 5$$

$$= 19 + 26i$$

ii) $\frac{-2 + 3i}{3 - 4i} \times \frac{3 + 4i}{3 + 4i} = \frac{-6 - 8i + 9i - 12}{9 + 16}$

$$= \frac{-18 + i}{25}$$

$$= -\frac{18}{25} + \frac{1}{25}i$$

c) $(x + iy)^2 = 9 + 40i$

$$x^2 - y^2 + 2xyi = 9 + 40i$$

equating

$$x^2 - y^2 = 9 \quad \text{--- (1)}$$

$$2xy = 40 \quad \text{--- (2)}$$

$$(x^2 + y^2)^2 = (x^2 - y^2)^2 + (2xy)^2$$

$$= 81 + 1600$$

$$= 1681$$

$$x^2 + y^2 = 41 \quad \text{--- (3)}$$

① + ③

$$2x^2 = 50$$

$$x^2 = 25$$

$$x = \pm 5$$

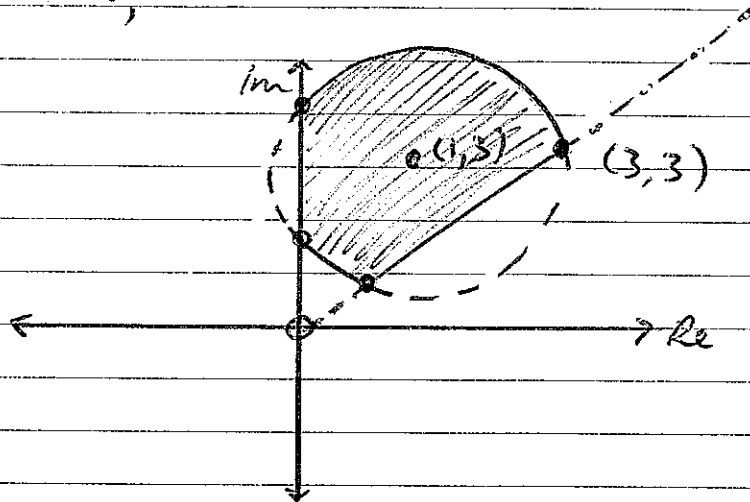
sub into ②

$$2(\pm 5)y = 40$$

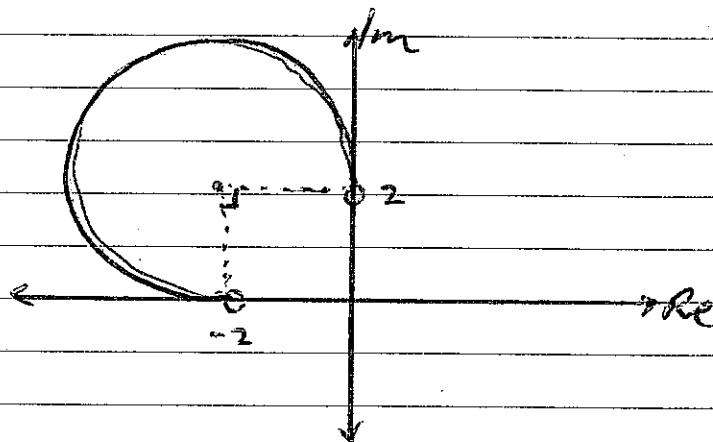
$$y = \pm 4$$

$$\therefore 5+4i, -5-4i$$

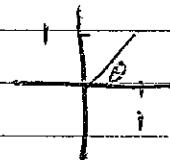
d) i)



ii)



e) i) $1+i = \sqrt{2} \left(\cos \frac{\pi}{4} + i \sin \frac{\pi}{4} \right)$



ii) $(1+i)^n = x+iy$

$$\left(\sqrt{2} \left(\cos \frac{\pi}{4} + i \sin \frac{\pi}{4} \right) \right)^n = x+iy$$

$$\left(2^{\frac{1}{2}} \right)^n \left(\cos n \frac{\pi}{4} + i \sin n \frac{\pi}{4} \right) = x+iy$$

$$\left| 2^{\frac{n}{2}} \left(\cos n \frac{\pi}{4} + i \sin n \frac{\pi}{4} \right) \right| = |x+iy|$$

$$2^{\frac{n}{2}} = \sqrt{x^2 + y^2}$$

$$x^2 + y^2 = 2^n$$

f) complex numbers with a modulus of 1.

consider $z = \frac{1}{z}$

$$z\bar{z} = 1$$

$$|z|^2 = 1$$

$$|z| = 1$$

g) $y = 2 \cos^{-1}(x^2 - 1)$

i) D: $-1 \leq x^2 - 1 \leq 1$
 $0 \leq x^2 \leq 2$
 $-\sqrt{2} \leq x \leq \sqrt{2}$

R: $x^2 \geq 0$
 $x^2 - 1 \geq -1$

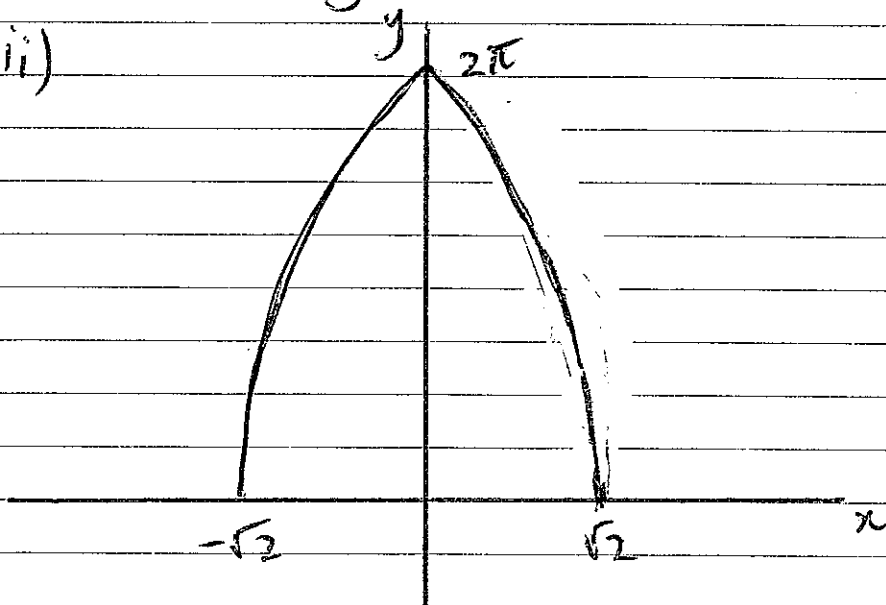
since $x^2 - 1$ will give all values between -1 & 1

$$0 \leq \cos^{-1}(x^2 - 1) \leq \pi$$

$$0 \leq 2 \cos^{-1}(x^2 - 1) \leq 2\pi$$

$$0 \leq y \leq 2\pi$$

ii)



$$\text{iii)} \quad y = 2\cos^{-1}(x^2-1)$$

$$y' = 2 \times \frac{-1}{\sqrt{1-(x^2-1)^2}} \times 2x$$

$$= \frac{-4x}{\sqrt{1-(x^4-2x^2+1)}}$$

$$= \frac{-4x}{\sqrt{2x^2-x^4}}$$

OR

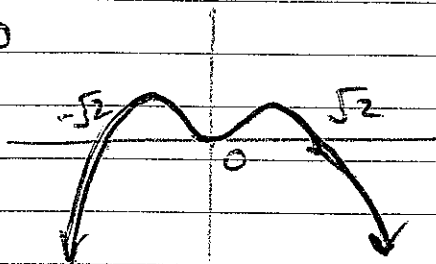
$$= \begin{cases} \frac{4}{\sqrt{2-x^2}}, & \text{when } x > 0 \\ \frac{4}{\sqrt{2-x^2}}, & \text{when } x < 0 \end{cases}$$

The derivative is defined when

$$2x^2 - x^4 > 0$$

$$x^2(2-x^2) > 0$$

$$x^2(\sqrt{2}-x)(\sqrt{2}+x) > 0$$



$$-\sqrt{2} < x < 0, \quad 0 < x < \sqrt{2}$$

OR

$$-\sqrt{2} < x < \sqrt{2}, \quad x \neq 0$$

Note: as $x \rightarrow 0^+$, $y' \rightarrow -2\sqrt{2}$

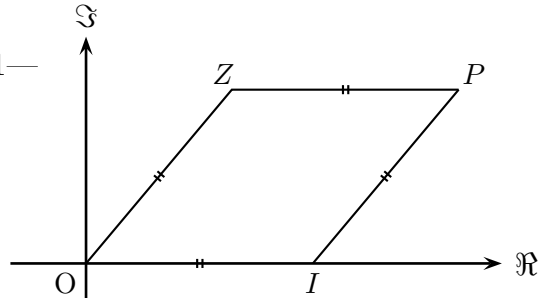
as $x \rightarrow 0^-$, $y' \rightarrow 2\sqrt{2}$

2014 Extension 2 Mathematics Task 1:
Solutions— Question 2

4

2. (a) The points O , I , Z , and P on the Argand diagram represent the complex numbers 0 , 1 , z , and $z + 1$ respectively, where $z = \cos \theta + i \sin \theta$ is any complex number of modulus 1 , and $0 < \theta < \pi$.
 (i) Explain why $OIPZ$ is a rhombus.

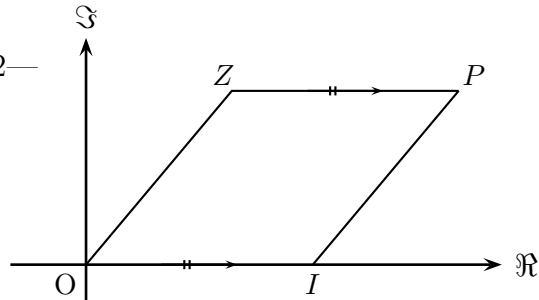
Solution: Method 1—



$$\begin{aligned}
 |OI| &= 1, & |OZ| &= 1, \\
 |ZP| &= |z + 1 - z|, & |IP| &= |z + 1 - 1|, \\
 &= |1|, & &= |z|, \\
 &= 1. & &= 1.
 \end{aligned}$$

$\therefore OIPZ$ is a rhombus (equal sides).

Solution: Method 2—



$$\begin{aligned}
 |OI| &= |ZP| = 1 \text{ by construction,} \\
 OI &\parallel ZP, \\
 \therefore OIPZ &\text{ is a parallelogram (opp. sides equal and parallel),} \\
 |OI| &= |OZ| = 1 \text{ (given),} \\
 \therefore OIPZ &\text{ is a rhombus.}
 \end{aligned}$$

- (ii) Show that $\frac{z-1}{z+1}$ is purely imaginary.

Solution: Method 1—

Consider the diagonals of the rhombus $OIPZ$:

$$\begin{aligned}
 OP &= z + 1, \\
 IZ &= z - 1, \\
 \arg(z - 1) - \arg(z + 1) &= \frac{\pi}{2}, \text{ (} OP \perp IZ, \text{ diagonals of rhombus)} \\
 \text{i.e., } \arg\left(\frac{z-1}{z+1}\right) &= \frac{\pi}{2}.
 \end{aligned}$$

So $\frac{z-1}{z+1}$ must lie on the imaginary axis and is purely imaginary.

Solution: Method 2—

$$\begin{aligned}\frac{z-1}{z+1} \times \frac{\bar{z}+1}{\bar{z}+1} &= \frac{z\bar{z}+z-\bar{z}-1}{z\bar{z}+z+\bar{z}+1}, \\ &= \frac{1+2i\sin\theta-1}{1+2\cos\theta+1}, \\ &= \frac{2i\sin\theta}{2+2\cos\theta}, \\ &= \frac{i\sin\theta}{1+\cos\theta}, \text{ which is purely imaginary.}\end{aligned}$$

Solution: Method 3—

If $\frac{z-1}{z+1}$ is purely imaginary, then $\frac{z-1}{z+1} + \overline{\left(\frac{z-1}{z+1}\right)} = 0$.

$$\begin{aligned}\text{L.H.S.} &= \frac{z-1}{z+1} + \frac{\bar{z}-1}{\bar{z}+1}, \\ &= \frac{z\bar{z}+z-\bar{z}-1+z\bar{z}+\bar{z}-z-1}{z\bar{z}+z+\bar{z}+1}.\end{aligned}$$

$$\text{But } z\bar{z} = |z|^2 = 1,$$

$$\begin{aligned}\text{so L.H.S.} &= \frac{0}{z+\bar{z}+2}, \\ &= 0, \\ &= \text{R.H.S.}\end{aligned}$$

Solution: Method 4—

$$\begin{aligned}\frac{\cos\theta+i\sin\theta-1}{\cos\theta+i\sin\theta+1} \times \frac{\cos\theta-i\sin\theta+1}{\cos\theta-i\sin\theta+1} &= \frac{\cos^2\theta-i\sin\theta\cos\theta+\cos\theta+i\sin\theta\cos\theta+\sin^2\theta+i\sin\theta-\cos\theta+i\sin\theta-1}{\cos^2\theta-i\sin\theta\cos\theta+\cos\theta+i\sin\theta\cos\theta+\sin^2\theta+i\sin\theta+\cos\theta-i\sin\theta+1} \\ &= \frac{2i\sin\theta}{2+2\cos\theta}, \\ &= \frac{i\sin\theta}{1+\cos\theta}, \text{ which is purely imaginary.}\end{aligned}$$

Solution: Method 5—

$$\begin{aligned}\frac{x-1+iy}{x+1+iy} \times \frac{x+1-iy}{x+1-iy} &= \frac{x^2+x-ixy-x-1+iy+ixy+iy+y^2}{(x+1)^2+y^2}, \\ &= \frac{x^2+y^2-1+2iy}{(x+1)^2+y^2}.\end{aligned}$$

$$\text{But } x^2+y^2 = 1 \text{ (i.e. } |z|^2),$$

$$\text{so } \frac{z-1}{z+1} = \frac{2iy}{(x+1)^2+y^2}, \text{ which is purely imaginary.}$$

Solution: Method 6—

$$\begin{aligned}
 \frac{z-1}{z+1} &= \frac{\cos \theta + i \sin \theta - 1}{\cos \theta + i \sin \theta + 1}, \\
 &= \frac{1 - 2 \sin^2 \frac{\theta}{2} + 2i \sin \frac{\theta}{2} \cos \frac{\theta}{2} - 1}{2 \cos^2 \frac{\theta}{2} - 1 + 2i \sin \frac{\theta}{2} \cos \frac{\theta}{2} + 1}, \\
 &= \frac{-2 \sin \frac{\theta}{2} (\sin \frac{\theta}{2} - i \cos \frac{\theta}{2})}{2 \cos \frac{\theta}{2} (\cos \frac{\theta}{2} + i \sin \frac{\theta}{2})}, \\
 &= \frac{i \sin \frac{\theta}{2} (\cos \frac{\theta}{2} + i \sin \frac{\theta}{2})}{\cos \frac{\theta}{2} (\cos \frac{\theta}{2} + i \sin \frac{\theta}{2})} \quad (\text{as } -1 = i^2), \\
 &= i \tan \frac{\theta}{2} \text{ which is purely imaginary.}
 \end{aligned}$$

(iii) Find the modulus of $z + 1$ in terms of θ .

Solution:

$$\begin{aligned}
 |z + 1|^2 &= (z + 1)(\bar{z} + 1), \\
 &= 2 + 2 \cos \theta \text{ as above,} \\
 \therefore |z + 1| &= \sqrt{2(1 + \cos \theta)}, \\
 &= \sqrt{2 \times 2 \cos^2 \frac{\theta}{2}}, \\
 &= 2 \cos \frac{\theta}{2}.
 \end{aligned}$$

(b) Differentiate $x \sin 2x$, and hence find $\int x \cos 2x \, dx$.

2

Solution:

$$\begin{aligned}
 \frac{d}{dx}(x \sin 2x) &= \sin 2x + 2x \cos 2x, \\
 \text{i.e., } 2x \cos 2x &= \frac{d}{dx}(x \sin 2x) - \sin 2x. \\
 \int 2x \cos 2x \, dx &= x \sin 2x - \int \sin 2x \, dx, \\
 &= x \sin 2x + \frac{\cos 2x}{2} + C. \\
 \text{So } \int x \cos 2x \, dx &= \frac{x \sin 2x}{2} + \frac{\cos 2x}{4} + C. \\
 \text{Alternatively, } \int 2x \cos 2x \, dx &= x \sin 2x - \int 2 \sin x \cos x \, dx, \\
 &= x \sin 2x - \sin^2 x + C. \\
 \text{So } \int x \cos 2x \, dx &= \frac{x \sin 2x - \sin^2 x}{2} + C.
 \end{aligned}$$

(c) Given that $2 - i$ is a root of the equation $x^4 - 6x^3 + 10x^2 + 2x - 15 = 0$:

5

(i) state another complex (non-real) root, giving a reason.

Solution: $2 + i$, as polynomials with real coefficients have their complex roots occurring in conjugate pairs.

(ii) find all the roots of the equation.

Solution: Method 1—

Possible other roots are $\pm 1, \pm 3, \pm 5$.

$$\begin{aligned} P(1) &= 1 - 6 + 10 + 2 - 15, \\ &\neq 0. \end{aligned}$$

$$\begin{aligned} P(-1) &= 1 + 6 + 10 - 2 - 15, \\ &= 0. \end{aligned}$$

$$\begin{aligned} P(3) &= 81 - 162 + 90 + 6 - 15, \\ &= 0. \end{aligned}$$

$\therefore$ The roots are $2 \pm i, -1$, and 3 .

Solution: Method 2—

$$\begin{aligned} (x - 2 - i)(x - 2 + i) &= x^2 - 4x + 4 + 1, \\ &= x^2 - 4x + 5. \end{aligned}$$

$$\begin{array}{r} x^2 - 4x + 5 \quad \overline{) \quad x^4 - 6x^3 + 10x^2 + 2x - 15} \\ \underline{-x^4 + 4x^3 - 5x^2} \\ -2x^3 + 5x^2 + 2x \\ \underline{2x^3 - 8x^2 + 10x} \\ -3x^2 + 12x - 15 \\ \underline{3x^2 - 12x + 15} \\ 0 \end{array}$$

$$x^2 - 2x - 3 = (x - 3)(x + 1)$$

$\therefore$ The roots are $2 \pm i, -1$, and 3 .

(iii) write the equation in fully factored form over the complex field.

Solution: $(x + 1)(x - 3)(x - 2 - i)(x - 2 + i) = 0$.

(d) Consider the functions $y = -\cos^{-1}\left(\frac{x}{2}\right)$ and $y = \frac{1}{2}\tan^{-1}(x) - \frac{\pi}{2}$.

4

(i) Show that the graphs of these functions intersect on the y -axis.

Solution: For $y = -\cos^{-1}\left(\frac{x}{2}\right)$, Domain : $-1 \leq \frac{x}{2} \leq 1$,

$$-2 \leq x \leq 2.$$

$$\text{Range : } -\pi \leq y \leq 0.$$

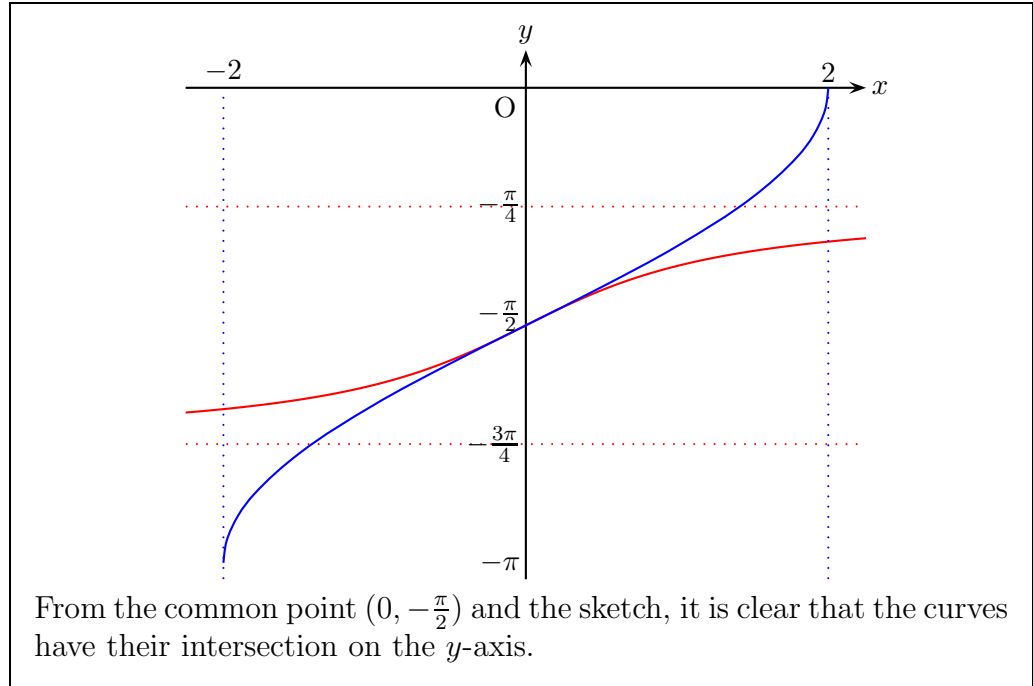
$$\text{When } y = 0, \quad x = -\frac{\pi}{2}.$$

For $y = \frac{1}{2}\tan^{-1}(x) - \frac{\pi}{2}$, Domain : $x \in \mathbb{R}$,

$$\text{Range : } -\frac{\pi}{4} - \frac{\pi}{2} < y < \frac{\pi}{4} - \frac{\pi}{2},$$

$$-\frac{3\pi}{4} < y < -\frac{\pi}{4},$$

$$\text{When } y = 0, \quad x = -\frac{\pi}{2}.$$



- (ii) Show that these graphs have a common tangent at the point of intersection, and write the equation of this tangent.

Solution:

$$y = -\cos^{-1}\left(\frac{x}{2}\right), \quad y = \frac{1}{2}\tan^{-1}(x) - \frac{\pi}{2},$$

$$\frac{dy}{dx} = -\frac{1}{2} \times \frac{-1}{\sqrt{1 - \frac{x^2}{4}}}, \quad \frac{dy}{dx} = \frac{1}{2} \times \frac{1}{x^2 + 1},$$

$$= \frac{1}{\sqrt{4 - x^2}}.$$

When $x = 0$, $\frac{dy}{dx} = \frac{1}{2}$. When $x = 0$, $\frac{dy}{dx} = \frac{1}{2}$.

$\therefore$ The tangents have a common slope and a common point, *i.e.*, a common tangent.

$$y - \left(-\frac{\pi}{2}\right) = \frac{1}{2}(x - 0),$$

$$2y + \pi = x,$$

$$x - 2y - \pi = 0 \text{ is the equation of the common tangent.}$$

- (e) Given the quadratic equation $x^2 - x - 3 = 0$ with roots α_1, α_2 :

5

- (i) Show that $x^4 = 7x + 12$.

Solution:

$$\begin{aligned} x^2 &= x + 3, \\ x^4 &= x^2 + 6x + 9, \\ &= (x + 3) + 6x + 9, \\ &= 7x + 12. \end{aligned}$$

- (ii) Hence or otherwise find a quadratic equation with roots α_1^4 and α_2^4 .

Solution: Method 1—

$$\text{Put } y = x^4, \text{ i.e., } x = y^{1/4},$$

$$y = 7y^{1/4} + 12,$$

$$y^{1/4} = \frac{y-12}{7}.$$

$$0 = \left(\frac{y-12}{7}\right)^2 - \frac{y-12}{7} - 3,$$

$$= y^2 - 24y + 144 - 7y + 84 - 147,$$

$$= y^2 - 31y + 81.$$

So the desired equation is $x^2 - 31x + 81 = 0$.

Solution: Method 2—

$$\alpha_1 + \alpha_2 = 1,$$

$$\alpha_1\alpha_2 = -3,$$

$$\alpha_1^4 = 7\alpha_1 + 12,$$

$$\alpha_2^4 = 7\alpha_2 + 12,$$

$$\alpha_1^4 + \alpha_2^4 = 7(\alpha_1 + \alpha_2) + 24,$$

$$= 7(1) + 24,$$

$$= 31.$$

$$\alpha_1^4\alpha_2^4 = 49\alpha_1\alpha_2 + 84(\alpha_1 + \alpha_2) + 144,$$

$$= 49(-3) + 84(1) + 144,$$

$$= 81.$$

$$\therefore x^2 - 31x + 81 = 0.$$

Solution: Method 3—

$$\alpha_1 + \alpha_2 = 1,$$

$$\alpha_1\alpha_2 = -3,$$

$$(\alpha_1 + \alpha_2)^2 = \alpha_1^2 + 2\alpha_1\alpha_2 + \alpha_2^2 = 1,$$

$$\alpha_1^2 + \alpha_2^2 = 1 - 2(-3),$$

$$= 7,$$

$$\alpha_1^2\alpha_2^2 = 9,$$

$$(\alpha_1^2 + \alpha_2^2)^2 = \alpha_1^4 + 2\alpha_1^2\alpha_2^2 + \alpha_2^4 = 49,$$

$$\alpha_1^4 + \alpha_2^4 = 49 - 2(9),$$

$$= 31,$$

$$\alpha_1^4\alpha_2^4 = 81,$$

$$\therefore x^2 - 31x + 81 = 0.$$

Solution: Method 4—

$$\begin{aligned}\text{Put } y &= x^4, \text{ i.e., } x = y^{1/4}, \\ (y^{1/4})^2 - y^{1/4} - 3 &= 0, \\ y^{1/4} &= y^{1/2} - 3, \\ (y^{1/4})^2 &= (y^{1/2} - 3)^2, \\ y^{1/2} &= y - 6y^{1/2} + 9, \\ (7y^{1/2})^2 &= (y + 9)^2, \\ 49y &= y^2 + 18y + 81, \\ 0 &= y^2 - 31y + 81.\end{aligned}$$

So the desired equation is $x^2 - 31x + 81 = 0$.

Solution: Method 5—

$$\begin{aligned}\alpha^2 - \alpha + \frac{1}{4} &= 3 + \frac{1}{4}, \\ \left(\alpha - \frac{1}{2}\right)^2 &= \frac{13}{4}, \\ \alpha - \frac{1}{2} &= \pm \frac{\sqrt{13}}{2}, \\ \alpha &= \frac{1 \pm \sqrt{13}}{2}, \\ \alpha^2 &= \frac{1 \pm 2\sqrt{13} + 13}{4}, \\ &= \frac{14 \pm 2\sqrt{13}}{4}, \\ &= \frac{7 \pm \sqrt{13}}{2}, \\ \alpha^4 &= \frac{49 \pm 14\sqrt{13} + 13}{4}, \\ &= \frac{62 \pm 14\sqrt{13}}{4}, \\ &= \frac{31 \pm 7\sqrt{13}}{2}, \\ \alpha_1^4 + \alpha_2^4 &= 31, \\ \alpha_1^4 \alpha_2^4 &= \frac{31^2 - 49 \times 13}{4}, \\ &= 81, \\ \therefore x^2 - 31x + 81 &= 0.\end{aligned}$$

QUESTION 3

(a) (i) $z^5 = 1$

$$z_0 = \text{cis } 0 = 1$$

$$z_1 = \text{cis } \frac{2\pi}{5}$$

2

$$z_2 = \text{cis } \frac{4\pi}{5}$$

$$z_3 = \text{cis } -\frac{2\pi}{5} = \overline{z_1}$$

$$z_4 = \text{cis } -\frac{4\pi}{5} = \overline{z_2}$$

(ii) $(z-1)(z-z_1)(z-\overline{z_1})(z-z_2)(z-\overline{z_2}) = z^5 - 1$ 2

$$(z^2 - 2z \cos \frac{2\pi}{5} + 1)(z^2 - 2z \cos \frac{4\pi}{5} + 1) = z^4 + z^3 + z^2 + z + 1.$$

(iii) coeff of z

$$-2 \cos \frac{2\pi}{5} - 2 \cos \frac{4\pi}{5} = 1.$$

1

$$\cos \frac{2\pi}{5} + \cos \frac{4\pi}{5} = -\frac{1}{2}$$

$$\cos 2A = 2\cos^2 A - 1$$

$$\cos \frac{2\pi}{5} + 2\cos^2 \frac{2\pi}{5} - 1 = -\frac{1}{2}.$$

$$4\cos^2 \frac{2\pi}{5} + 2\cos \frac{2\pi}{5} - 1 = 0.$$

$$\text{Let } u = \cos \frac{2\pi}{5}$$

$$4u^2 + 2u - 1 = 0$$

$$(u + \frac{1}{4})^2 - \frac{1}{16} = \frac{4}{16}$$

$$u = -\frac{1}{4} \pm \frac{\sqrt{5}}{4}$$

2

$$\cos \frac{2\pi}{5} = \frac{\sqrt{5}-1}{4}$$

since $\frac{2\pi}{5}$ is in the first quadrant.

(b)

$$p(x) = A(x)Q(x) + (ax+b)$$

$$p(x) = (x-2)(x-3)Q(x) + (ax+b)$$

$$p(2) = 2a+b = 4 \quad (1)$$

$$p(3) = 3a+b = 9 \quad (2)$$

2

$$(2) - (1)$$

$$a = 5$$

$$10 + b = 4$$

$$b = -6$$

remainder is $5x-6$

$$(c) (i) 9!$$

1

$$(ii) 4! \times 2 \times 3 \times 8 = 1152$$

or

$$2^3 \times 3! \times {}^4C_3 \times 3! = 1152$$

2

$$\frac{1152}{9!} = \frac{1}{315}$$

$$(d) \quad ax^4 + bx + c = 0$$

If this has a triple root then

$$4ax^3 + b = 0$$

has a double root. So

$$12ax^2 = 0$$

has a single root, which is,

$$x = 0.$$

But $x = 0$ is not a root of $ax^4 + bx + c = 0$.

$$(e) \quad \int_1^{\sqrt{3}} \frac{x^2}{\sqrt{4-x^2}} dx$$

$$x = 2 \sin \theta$$

$$\frac{dx}{d\theta} = 2 \cos \theta.$$

$$dx = 2 \cos \theta d\theta.$$

$$\begin{array}{ccc} x & & \theta \\ \sqrt{3} & \rightarrow & \frac{\pi}{3} \\ 1 & \rightarrow & \frac{\pi}{6} \end{array}$$

$$\int_{\frac{\pi}{6}}^{\frac{\pi}{3}} \frac{4 \sin^2 \theta \cdot 2 \cos \theta}{\sqrt{4 - 4 \sin^2 \theta}} d\theta$$

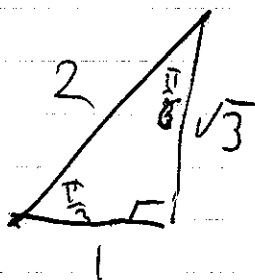
$$= 4 \int_{\frac{\pi}{6}}^{\frac{\pi}{3}} \frac{\sin^2 \theta \cos \theta}{\sqrt{1 - \sin^2 \theta}} d\theta$$

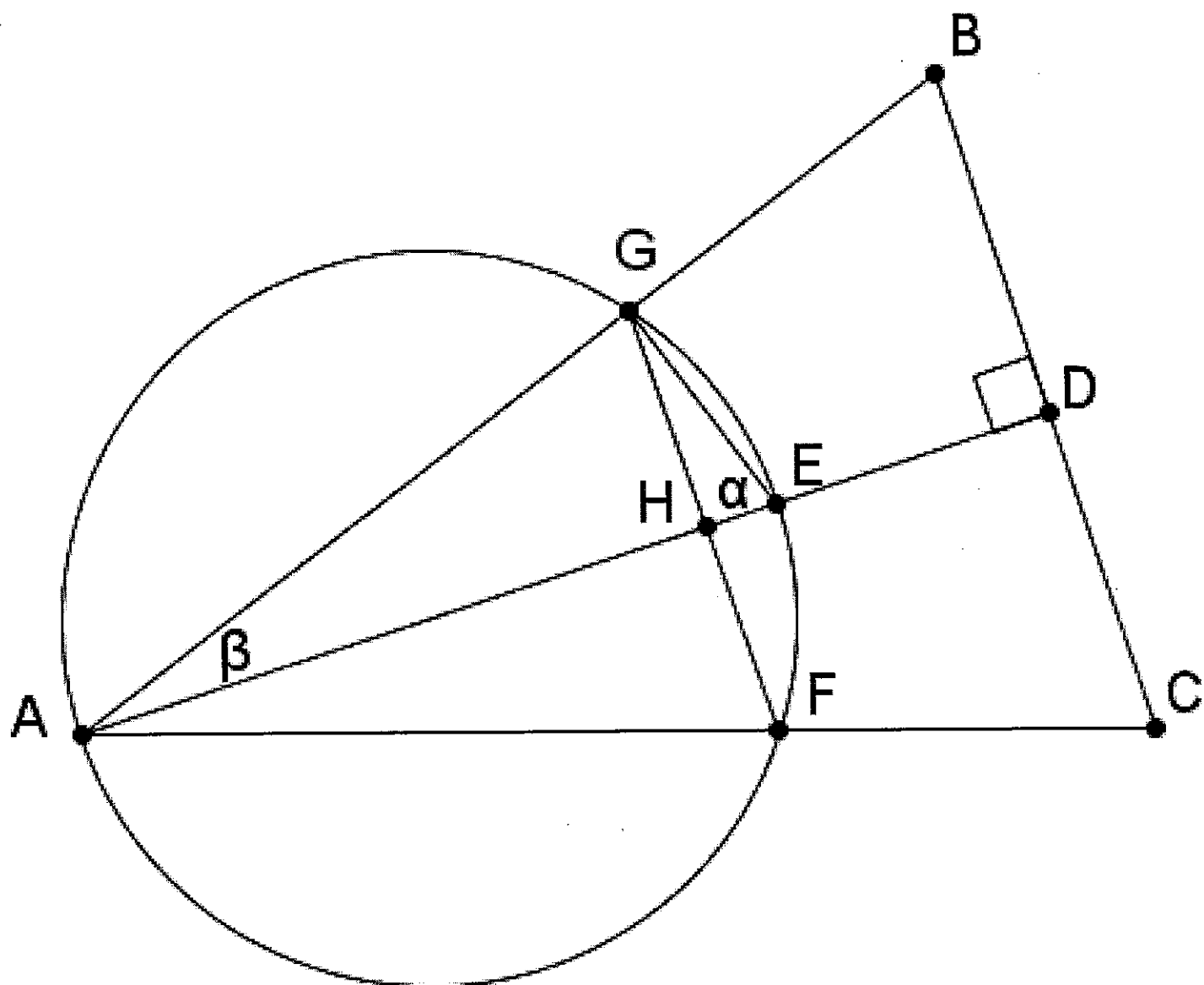
$$= 4 \int_{\frac{\pi}{6}}^{\frac{\pi}{3}} \sin^2 \theta d\theta$$

$$= 2 \int_{\frac{\pi}{6}}^{\frac{\pi}{3}} 1 - \cos 2\theta d\theta$$

$$= 2 \left[\theta - \frac{\sin 2\theta}{2} \right]_{\frac{\pi}{6}}^{\frac{\pi}{3}}$$

$$= 2 \left[\left(\frac{\pi}{3} - \frac{\sin 2\frac{\pi}{3}}{2} \right) - \left(\frac{\pi}{6} - \frac{\sin \frac{\pi}{3}}{2} \right) \right]$$





$$= 2 \left[\frac{\pi}{3} - \frac{\pi}{6} - \sin \frac{\pi}{3} \cos \frac{\pi}{3} + \frac{\sin \frac{\pi}{3}}{2} \right]$$

$$= \frac{\pi}{3} - 2 \times \frac{\sqrt{3}}{2} \times \frac{1}{2} + \frac{\sqrt{3}}{2}$$

$$= \frac{\pi}{3}$$

3

(f) Join GF. Let H be the intersection of GF and AE.

Join GE

Let $\angle GEA = \alpha^\circ$ and $\angle GAE = \beta^\circ$

$\triangle AGE$ is a right angle triangle with $\angle AGE = 90^\circ$ (angle in a semi-circle)

$\therefore \alpha + \beta = 90^\circ$ ($\angle$ sum $\triangle$)

Since $\triangle ADB$ is right angle triangle $\angle ABD = \alpha^\circ$ ($\angle$ sum of a $\triangle$).

$\angle GFA = \alpha^\circ$ ($\angle$ s in the same segment).

$\angle GFC = 180^\circ - \alpha^\circ$ (supplementary).

$\therefore BCFG$ is cyclic (opposite $\angle$ s are supplementary)

4